

IS 269:2015 — Ordinary Portland Cement Specification

Structured reference sheet: key tables and numeric parameters, summarized for technical/compliance-checking use.

Note: This is a working summary of published requirements, not a reproduction of the full IS 269:2015 standard text. Refer to the original BIS document (available for purchase from BIS) for complete, authoritative clauses.

Scope (summary)

IS 269:2015 (Sixth Revision) specifies requirements for 43-grade ordinary Portland cement (OPC), covering manufacturing constituents, permitted performance improvers/mineral admixtures, chemical composition limits, and physical performance requirements (strength, setting time, fineness, soundness) across grades OPC 33, 43, 43S, 53 and 53S.

Table 1 — Permitted Performance Improvers (Mineral Admixtures)

Max addition: 5% by mass for each material listed, subject to the stated conformity requirements.

Sl No.	Performance Improver	Max % by Mass	Requirement
i)	Fly ash	5	Conforming to IS 3812 (Part 1)
ii)	Granulated slag	5	Conforming to IS 12089
iii)	Silica fume	5	Conforming to IS 15388
iv)	Limestone	5	CaCO ₃ content (calculated from CaO content) not less than 75%, tested as per IS 1760 (Part 3)
v)	Rice husk ash	5	a) Reactive silica not less than 80%, tested as per IS 3812 (Part 1) b) Pozzolanic activity index not less than 90%, tested as per 10 of IS 1727 c) Loss on ignition not more than 5.0%, tested as per IS 1727
vi)	Metakaolin	5	Conforming to IS 16354
vii)	Copper slag	5	a) Lumps exceeding 50 mm: not more than 5% of mass of slag b) Chemical composition per IS 4032: SiO ₂ : 20-40% Fe ₂ O ₃ : 40-70% SO ₃ Max: 1% c) Glass content (optical microscope method): not less than 25%
viii)	Steel slag	5	a) Lumps exceeding 10 mm: not more than 5% of mass of slag b) Chemical composition per IS 4032: CaO: 40-55% SiO ₂ : 10-20% MgO: 1-5% MnO ₂ : 0-1% Al ₂ O ₃ : 0.5-4% Fe(total): 15-30% P: 0.5-2% c) Glass content: not less than 25%
ix)	Lead-Zinc slag	5	a) Lumps exceeding 10 mm: not more than 5% of mass of slag b) Chemical composition per IS 4032: ZnO: 5-15% PbO: 0.5-2% Fe ₂ O ₃ : 30-40% SiO ₂ : 15-25% CaO: 10-20% Al ₂ O ₃ : 10-15% MgO: 1-3% c) Glass content: not less than 25%
x)	Spent fluidized catalytic cracking equilibrium	5	Conforming to IS 1344

Sl No.	Performance Improver	Max % by Mass	Requirement
	catalyst		

Table 2 — Chemical Requirements (by grade)

Sl No.	Characteristic	OPC 33	OPC 43	OPC 43S	OPC 53	OPC 53S
i)	Ratio: lime to (silica, alumina, iron oxide) — LSF formula	0.66-1.02	0.66-1.02	0.80-1.02	0.80-1.02	0.80-1.02
ii)	Ratio: alumina to iron oxide, Min	0.66	0.66	0.66	0.66	0.66
iii)	Insoluble residue, % by mass, Max	5.0	5.0	2.0	5.0	2.0
iv)	Magnesia, % by mass, Max	6.0	6.0	5.0	6.0	5.0
v)	Total sulphur (as SO ₃), % by mass, Max	3.5	3.5	3.5	3.5	3.5
vi)	Loss on ignition, % by mass, Max	5.0	5.0	4.0	4.0	4.0
vi i)	Chloride content, % by mass, Max	0.1 (0.05 prestressed)	0.1 (0.05 prestressed)	0.1 (0.05 prestressed)	0.1 (0.05 prestressed)	0.1 (0.05 prestressed)
vi ii)	Alkali content — see Note 1 (optional, when specified)	-	-	-	-	-

Table 3 — Physical Requirements (by grade)

Sl No.	Characteristic	OPC 33	OPC 43	OPC 43S	OPC 53	OPC 53S	Test Method
i)	Fineness, m ² /kg, Min	225	225	370	225	370	IS 4031 (Part 2)
ii) a)	Soundness — Le-Chatelier, mm, Max	10	10	5	10	5	IS 4031 (Part 3)
ii) b)	Soundness — Autoclave test, %, Max	0.8	0.8	0.8	0.8	0.8	IS 4031 (Part 3)
iii) a)	Setting time — Initial, min, Min	30	30	60	30	60	IS 4031 (Part 5)
iii) b)	Setting time — Final, min, Max	600	600	600	600	600	IS 4031 (Part 5)
iv)	Compressive strength, 72±1h,	16	23	23	27	27	IS 4031 (Part 6)

Sl N o.	Characteristic	OPC 33	OPC 43	OPC 43S	OPC 53	OPC 53S	Test Method
iv)a)	MPa, Min						
iv)b)	Compressive strength, 168±2h, MPa, Min	22	33	37.5	37	37.5	IS 4031 (Part 6)
iv)c)	Compressive strength, 672±4h, MPa, Min	33	43	43	53	53	IS 4031 (Part 6)
iv)	Compressive strength, 672±4h, MPa, Max	48	58	-	-	-	IS 4031 (Part 6)
v)	Transverse strength (optional)	See Notes	See Notes	See Notes	See Notes	See Notes	IS 4031 (Part 8)

Key Reference Standards Cited

IS 4031 (Parts 2,3,5,6,8) — Test methods (fineness, soundness, setting time, strength, transverse strength)

IS 4032 — Chemical analysis method for slag composition

IS 3812 (Part 1) — Fly ash / reactive silica testing

IS 1727 — Pozzolanic activity index / loss on ignition testing

IS 12089 — Granulated slag specification

IS 15388 — Silica fume specification

IS 16354 — Metakaolin specification

IS 1344 — Spent fluidized catalytic cracking equilibrium catalyst

IS 1760 (Part 3) — Limestone CaCO₃ content testing